Electricity for the Gaggia

Electricity for the Gaggia, just enough

Live: https://rnaidenov.github.io/deep-dive-briefings/electricity-gaggia/

The short version

Three things to keep

1. FactCurrent only flows around a **closed loop**. A switch, a thermostat or a jumper just opens or closes that loop. S3
2. FactEU colours: **brown** = **Line**, **blue** = **Neutral**, **green/yellow** = **Earth**. S1 S2
3. FactGaggiMate's first safety step is: **unplug the machine** so nothing is live while you work. S4

First principles

Loops, switches, and the fault path

Fact A circuit is a closed loop that gives current a return path; without the loop it is an **open circuit** and nothing flows. S3

In the Gaggia, current leaves on Line (brown), goes through a part (heater, pump, valve), and comes back on Neutral (blue). Every switch and thermostat in between is just a gate in that loop.

Fact Earth is **not** the working path. It ties exposed metal to a low-impedance route so that if a live wire touches the chassis, a big fault current flows and trips the breaker/RCD or blows a fuse. S5

What “hot” meansPlain “Hot” or “live” is everyday wording for the Line conductor: the one at mains voltage relative to earth. In UK/EU terms both Line and Neutral count as live (current-carrying) conductors; earth does not. S2 Treat **both** brown and blue as dangerous while plugged in.

Colours

Brown, blue, green/yellow

1. Fact**Brown** — single AC phase (Line). IEC 60446/60445 colour scheme, which EU wiring is harmonised with. S1 S2
2. Fact**Blue** — Neutral. S1
3. Fact**Green/yellow** — always and only the protective (earth) conductor. S1

Colour is not identity inside the machineOpinion The internal Gaggia wires (grey, white, black, the kit’s red/green/yellow) are not mains-standard colours. GaggiMate has you **label by position** (BT.A, S.5, C.4, PMP.B…) for that reason; trust the label, not the colour. S4

Fact The cup-warmer has two **grounding wires** you disconnect to lift it off; GaggiMate says reconnect them on reassembly. S4 That is the earth bond for the metal chassis: don’t skip it.

Spades & jumpers

Why the red BT.A–BT.B jumper exists

Each spade connector is one point in a loop. Pull it off and the loop is open.

Fact The stock brew thermostat sits in series in the heater loop. GaggiMate has you remove it and **bridge BT.A and BT.B with the small red jumper**, securely and insulated. S4

Reasoning Removing the thermostat would leave the loop open, so the heater could never run. The jumper restores **continuity**, and temperature control moves to the thermocouple + board + SSR instead. See the Gaggia briefing → jumper.

High level

Three paths — don’t mix them up

1 · Heater → SSR

Fact ST.A/ST.B are extended with yellow wires to the **OUTPUT** side of the SSR; the yellow/black signal cable goes to its input. The SSR switches the **heater only**. S4 S6

2 · Pump → board P

Fact The kit’s green wire goes from the pump’s PMP.B position to the board’s **P/Pump** connector. S4 The pump is **not** on the SSR.

3 · Valve → board V

Fact The original PMP.B lead (grey; the one **not** going into the pump’s thermal fuse) is extended with the red wire to the board’s **V/Valve** connector. S4

Plus the board’s own feed: C.4 (brown) → **L/Line** via the grey kit wire; S.5 (white) → **N/Neutral** via the blue kit wire. S4 Where-is-what: BT / ST / PMP finder · 3D explore · companion.

Habits

Safe habits, every time

1. Fact**Unplug** at the wall before opening; work dry. S4
2. Opinion**Don’t probe live** mid-install. Test only after it’s closed up, per the guide’s final checks.
3. Fact**Never join earth to Line or Neutral**; green/yellow is exclusively protective. S1
4. Opinion**Leave mains brown/blue alone** unless the guide names that exact wire (e.g. C.4, S.5).
5. Fact**Label by position**, not colour, and keep a record of your wiring. S4
6. Fact**Insulate every join** and every removed end with tape or heatshrink. If unsure, ask a professional. S4

Receipts

Sources

All opened 29 Oct 2026. Official HD 308 / IEC 60445 texts are paywalled; S1 summarises the IEC colour scheme.

1. S1 IEC 60446 (now IEC 60445) — conductor coloursWebWikipedia Brown = single phase, blue = neutral, green/yellow exclusively protective.
2. S2 Electrical wiring in the United KingdomWebWikipedia L brown / N blue / E green-yellow; line and neutral are live conductors, earth is not; harmonised with Europe/IEC 60446.
3. S3 Electrical network / circuitWebWikipedia Circuit = closed loop with return path; no loop = open circuit.
4. S4 GaggiMate — Gaggia Classic Pro/Evo installationWebGaggiMate docs Unplug, high-voltage safety, ground wires, red BT jumper, SSR wiring, L/N/P/V connections, insulation.
5. S5 Ground (electricity)WebWikipedia Protective earth gives exposed metal a low-impedance fault path that trips breakers/RCDs.
6. S6 Solid-state relayWebWikipedia How an SSR switches an AC load electronically.